

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Vineesh Cutting
Roll Number	RSI2026505
Programme / Department	IT
Topic ID	T_33
Full Assigned Topic	Stylistic Homogenization Risk in Large Language Model-Assisted Academic Writing
AI Tool Used	Gemini Pro
Date	21-Aug-2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☒ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☒ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☒ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Gemini
Model name	Pro
Model/version shown	3.1
Access tier	☐ Free ☒ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☒ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☒ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☒ Yes ☐ No
Date/time generated	21-08-2026

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

*Generate a scholarly research paper on the topic: **"Stylistic Homogenization Risk in Large Language Model-Assisted Academic Writing"**. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Stylistic Homogenization Risk in Large Language Model-Assisted Academic Writing

Measure	Value
Approximate pages	4
Approximate word count	1492
Number of sections	9
Total number of references	4
Approximate in-text citation occurrences	8
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	4
PMID available, no DOI	
arXiv ID available, no DOI/PMID	
URL only	

No persistent identifier supplied	
TOTAL	4

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	NA
R2	NA
R3	NA
R4	NA

Gemini used an author-date citation style (like APA) for the in-text citations, such as (He et al., 2023).

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this **BEFORE** searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
------	--------------------	------------------	------------------

R1	Most info is given but citation is low.	4	Yes
R2	Journal name missing	3	Yes
R3	Volume Issue missing but high citations	4	Yes
R4	Journal name, volume, issue missing	3	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID
- ☐ Publisher or official repository
- ☐ Google Scholar
- ☐ Semantic Scholar / OpenAlex
- ☐ Crossref
- ☐ PubMed or another discipline-specific database
- ☐ Exact-title search
- ☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	N	Y	Y	N	N
R3	Y	Y	Y	Y	N	Y
R4	Y	N	Y	Y	N	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	ScienceDirect	DOI	Meta Data Accurately matched
R2	arXiv Indexer	DOI	Gemini referenced old paper while DOI navigated to New version of same paper
R3	arXiv Indexer	DOI	Meta Data Accurately matched
R4	arXiv Indexer	DOI	Gemini referenced old paper while DOI navigated to New version of same paper

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Gemini mentioned Info is correct
R2	E	Identifier opened different paper version
R3	A	Gemini mentioned Info is correct
R4	B	Title mismatch found.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = __2__ B = __1__ C = __0__ D = __0__ E = __1__

Total audited: N = A + B + C + D + E = __4__

Code	Points
A	4
B	3
C	1
D	0
E	1

$$\text{Authenticity Points} = 4A + 3B + C + E$$

$$\text{Authenticity Score} = [(4A + 3B + C + E) / (4N)] \times 100$$

$$\text{My calculation: } 4(\underline{2}) + 3(\underline{1}) + 1(\underline{0}) + 0(\underline{0}) + 1(\underline{1}) = \underline{12} \text{ points}$$

$$\text{Authenticity Score} = \underline{12} / (4 \times \underline{4}) \times 100 = \underline{75}$$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

$$\text{Prediction Accuracy} = (\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$$

$$\text{Correct predictions} = \underline{4} \quad \text{Total predictions} = \underline{4} \quad \text{Prediction Accuracy} = \underline{100} \%$$

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	4
References deeply audited	4
A: Verified	2
B: Wrong metadata	1
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	1
Authenticity Score	$\frac{75}{100}$
Prediction Accuracy	$\frac{100}{100} \%$

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Most of the citations were optimal yet some essential elements such as Volume, Issue, Page Number were missing.

.....

2. Did all genuine references actually support the claims for which they were cited?

Yes, most of the citations were feasible enough to support the paper generated by Gemini.

.....

3. What was the most serious citation-integrity problem you observed?

Lack of Bibliographical reference numbers, inconsistent citation format, outdated DOI references.

.....

4. What is the single most important lesson you learned from this lab?

AI can be an efficient collaborative partner, yet lack of proper or incomplete prompt can still lead to loss in accuracy therefore its highly recommended to act as quality control by critically reading, testing, and evaluating the AI's outputs, logic, and reasoning rather than accepting them blindly.

.....

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name:Vineesh Cutting.....

Roll Number:RSI2026505.....

Signature:N/A.....

Date:21-Aug-2026.....

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
Verification -> Quantitative Scores -> Submission

Stylistic Homogenization Risk in Large Language Model-Assisted Academic Writing

Abstract

The rapid integration of Large Language Models (LLMs) into scholarly workflows has fundamentally transformed academic writing, offering unprecedented efficiencies in drafting, literature summarization, and language refinement. However, this paradigm shift raises critical concerns regarding the preservation of epistemic and stylistic diversity in scientific discourse.

This paper examines the risk of stylistic homogenization—a phenomenon where overreliance on generative artificial intelligence (AI) converges academic writing into a standardized, algorithmic mono-voice. While LLMs act as a "linguistic equalizer" for non-native English speakers by elevating baseline lexical complexity, they concurrently risk diminishing the unique intellectual framing and creative originality of researchers. Through an analysis of current literature encompassing architectural foundations, automated review generation, and macroscopic lexical shifts, this paper investigates the dual-edged nature of LLM assistance. The study concludes by identifying key research gaps, notably the need for robust frameworks that balance automated linguistic enhancement with the preservation of diverse scholarly voices across multi-disciplinary domains.

Keywords: Large Language Models, Stylistic Homogenization, Academic Writing, Generative AI, Lexical Diversity, Scholarly Communication.

1. Introduction

The landscape of scholarly communication is undergoing a seismic technological transformation driven by the advent of Large Language Models (LLMs). Rooted in advanced transformer architectures and trained on massive textual corpora, these models have demonstrated remarkable, generalized capabilities across a multitude of natural language processing tasks (Naveed et al., 2023). Within the academic sphere, tools like ChatGPT are increasingly utilized to expedite the research process, assisting authors in outlining, drafting, and refining scientific manuscripts (He et al., 2023).

Despite these clear logistical advantages, the pervasive adoption of LLMs introduces complex ethical and cognitive challenges. Chief among these is the threat of stylistic homogenization. As millions of researchers globally rely on a centralized set of commercial algorithms to correct and shape their prose, the natural idiosyncrasies, disciplinary nuances, and creative expressions inherent to human authorship risk being flattened into a uniform algorithmic style. An overreliance on LLMs for content generation and ideation may diminish a researcher's capacity to formulate unique perspectives, ultimately yielding a stagnant academic writing style bereft of genuine originality (He et al., 2023). This paper critically evaluates this homogenization risk, exploring the tension between accessibility, efficiency, and the loss of stylistic diversity in contemporary academic writing.

2. Background, Related Work, and Literature Review

To understand the impact of LLMs on academic writing, it is essential to trace their architectural evolution and the scope of their capabilities. Early language models relied on task-specific supervised training; however, the paradigm shifted dramatically with the introduction of self-supervised, pre-trained transformer models scaled to hundreds of billions of parameters (Naveed et al., 2023). This scaling unlocked "emergent abilities," such as zero-shot learning, complex reasoning, and highly coherent text generation, allowing these models to execute complex academic tasks without domain-specific fine-tuning (Naveed et al., 2023).

The literature highlights a dichotomy in the reception of LLMs in academia. On one hand, researchers have successfully applied LLMs to automate complex phases of the writing process, such as reference generation, abstract writing, and literature review composition (Tang et al., 2025). On the other hand, empirical evaluations consistently reveal limitations. For instance, despite rapid algorithmic progress, advanced LLMs continue to suffer from factual hallucinations and semantic coverage deficits when attempting to synthesize comprehensive literature reviews, with their efficacy varying significantly across different academic disciplines (Tang et al., 2025). Consequently, current literature suggests that while LLMs are powerful assistive tools, they cannot safely operate autonomously without stringent human oversight.

3. Main Thematic Discussion: The Double-Edged Sword

The debate surrounding LLM-assisted academic writing is fundamentally characterized by the tension between linguistic equalization and stylistic homogenization.

3.1 LLMs as Linguistic Equalizers

A significant portion of global scientific literature is produced by Non-Native English Speakers (NNES), who historically face substantial linguistic barriers when targeting high-impact, English-dominant journals. Recent macroscopic analyses indicate that LLMs actively mitigate these barriers. By quantifying lexical shifts across millions of published abstracts, researchers have demonstrated that LLM adoption significantly enhances the lexical complexity of NNES-authored papers (Lin et al., 2025). In this capacity, LLMs serve as a vital "linguistic equalizer," reducing historical disparities and promoting a more equitable global academic landscape (Lin et al., 2025).

3.2 The Risk of Stagnation and Mono-Voice

Conversely, the widespread reliance on the same underlying statistical models introduces systemic risks to intellectual diversity. Because LLMs function fundamentally as next-token predictors optimized for fluency and statistical likelihood, they heavily favor centralized linguistic patterns over marginal, complex, or highly novel expressions. He et al. (2023) conceptualize this as a "double-edged sword": while the AI expedites mundane writing tasks and clarifies grammar, its excessive use can stifle original thought. When scholars outsource ideation and complex phrasing to LLMs, the resulting text often lacks the nuanced conceptual depth and creative friction that characterize rigorous scientific debate, threatening a broader homogenization of scientific thought (He et al., 2023).

4. Current Approaches or Developments

Recent developments in the study of AI-generated academic text have focused on quantifying exactly how these models alter the academic corpus.

1. **Macroscopic Lexical Tracking:** Utilizing sophisticated metrics such as the Measure of Textual Lexical Diversity (MTLD) and difference-in-differences (DID) causal designs, researchers are now mapping the precise shifts in vocabulary complexity occurring across global databases like OpenAlex (Lin et al., 2025). This approach provides empirical, causal evidence of how AI is structurally altering the vocabulary utilized in published abstracts.
2. **Automated Evaluation Frameworks:** As researchers increasingly use LLMs for literature reviews, new multi-dimensional evaluation frameworks have been developed to systematically score AI outputs. These frameworks measure not only lexical diversity but also factual consistency, natural language inference (NLI), and hallucination rates in AI-generated reference lists (Tang et al., 2025). These frameworks reveal that the utility of LLMs is heavily context-dependent and varies significantly based on the academic discipline being modeled.

5. Research Challenges and Limitations

The integration of generative AI into scholarly communication faces several profound limitations that researchers must navigate:

- **Hallucination and Verification:** A persistent technical limitation is the generation of plausible but entirely fictitious references. Evaluative frameworks demonstrate that even the most state-of-the-art models fail to completely eliminate hallucinated citations during automated literature review composition (Tang et al., 2025).
- **Erosion of Authorship and Accountability:** Because LLMs cannot be held legally or ethically accountable for their outputs, they violate foundational principles of academic authorship. Incorporating AI-generated prose without attribution challenges the integrity of the peer-review system and obscures the provenance of scientific claims (He et al., 2023).
- **The Epistemic Echo Chamber:** As LLMs generate more academic text, and future LLMs are trained on this very text, a feedback loop is created. This loop risks amplifying dominant writing styles and burying minority scholarly perspectives, compounding the homogenization effect.

6. Research Gaps and Future Directions

While the immediate lexical impacts of LLMs are becoming clearer, several critical research gaps remain:

1. **Longitudinal Cognitive Impact:** There is a pressing need for longitudinal studies examining how long-term reliance on LLMs affects the critical thinking and independent writing competencies of early-career researchers and graduate students.
2. **Disciplinary Nuance:** Current research indicates that LLM performance and stylistic influence vary across fields (Tang et al., 2025). Future research must investigate *why* certain disciplines (e.g., humanities vs. STEM) are more susceptible to AI-driven stylistic homogenization and how specific disciplinary registers can be preserved.
3. **Design of Stylistically Diverse AI:** Computer science research must prioritize the development of "tunable" LLMs that can reliably adopt diverse academic personas or specifically prioritize rare, novel syntactic structures to combat statistical homogenization (Naveed et al., 2023).

7. Conclusion

Large Language Models stand as one of the most disruptive technologies introduced to academic writing in the digital age (Naveed et al., 2023). Their capacity to act as a linguistic equalizer holds immense promise for democratizing global scientific communication, particularly for non-native English-speaking researchers (Lin et al., 2025). However, this benefit is inextricably linked to the severe risk of stylistic and cognitive homogenization. An overreliance on LLMs threatens to replace the vibrant, diverse ecosystem of human scholarly expression with a stagnant, homogenized algorithmic standard (He et al., 2023). To responsibly navigate this new era, the academic community must establish rigorous evaluation frameworks (Tang et al., 2025) and ethical guidelines that leverage AI for mechanical efficiency while fiercely protecting the irreplaceable creativity and stylistic diversity of the human intellect.

References

He, S., Yang, F., Zuo, J.-p., & Lin, Z.-m. (2023). ChatGPT for scientific paper writing—promises and perils. *The Innovation*, 4(100524), 100524. <https://doi.org/10.1016/j.xinn.2023.100524>

Cited by: 24

Lin, D., Zhao, N., Tian, D., & Li, J. (2025). ChatGPT as Linguistic Equalizer? Quantifying LLM-Driven Lexical Shifts in Academic Writing. *arXiv*. <https://doi.org/10.48550/arxiv.2504.12317>

Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., Akhtar, N., Barnes, N., & Mian, A. (2023). A Comprehensive Overview of Large Language Models. *arXiv*. <https://doi.org/10.48550/arxiv.2307.06435>

Cited by: 3510

Tang, X., Duan, X., & Cai, Z. G. (2025). Large Language Models for Automated Literature Review: An Evaluation of Reference Generation, Abstract Writing, and Review Composition. *arXiv*. <https://doi.org/10.48550/arxiv.2412.13612>

Cited by: 33